

JAVASCRIPT LIBRARY

# jQuery in JavaScript

Detailed Notes

— Write Less, Do More —

<> DOM Manipulation · Event Handling · Animations · AJAX



VERSION

3.x

# Introduction to jQuery



## DEFINITION

jQuery is a **fast, lightweight**, and **feature-rich** JavaScript library that simplifies HTML document traversal, event handling, animation, and AJAX interactions.

## KEY ADVANTAGE

Write **less code** while achieving **more functionality**

**Lightweight**

~30KB minified

**Cross-browser**

Compatible

**Easy**

To learn



# | Why jQuery is Used

## ↔ Code Comparison

### ● VANILLA JAVASCRIPT

```
document.getElementById("demo").  
style.color = "red";
```



### ● JQUERY

```
$("#demo").css("color", "red");
```



Shorter



Cleaner



Easier

## | Features of jQuery



01

### DOM Manipulation

Effortlessly select and modify HTML elements



02

### Event Handling

Simple and powerful event listeners



03

### Animations

Built-in animation effects and transitions



04

### AJAX Support

Seamless data loading without page refresh



05

### Cross-browser Compatibility

Works consistently across all browsers



06

### Simplified Programming

Cleaner syntax, less boilerplate code



# How to Add jQuery in a Web Page



## Method 1: Using CDN

RECOMMENDED

Load jQuery from a Content Delivery Network

```
<script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
```



## Method 2: Download jQuery

OFFLINE

Download jQuery library file and link locally

```
<script src="jquery.js"></script>
```



# | jQuery Syntax

## <> Basic Syntax

```
$(selector).action();
```



PART 1

jQuery Function



PART 2

Selects HTML Element



PART 3

Action to Perform



### 💡 Example

```
$("p").hide();
```

Hides all paragraph elements in the document

# | Document Ready Method



## Purpose

Ensures **HTML is fully loaded** before jQuery runs



## Standard Syntax

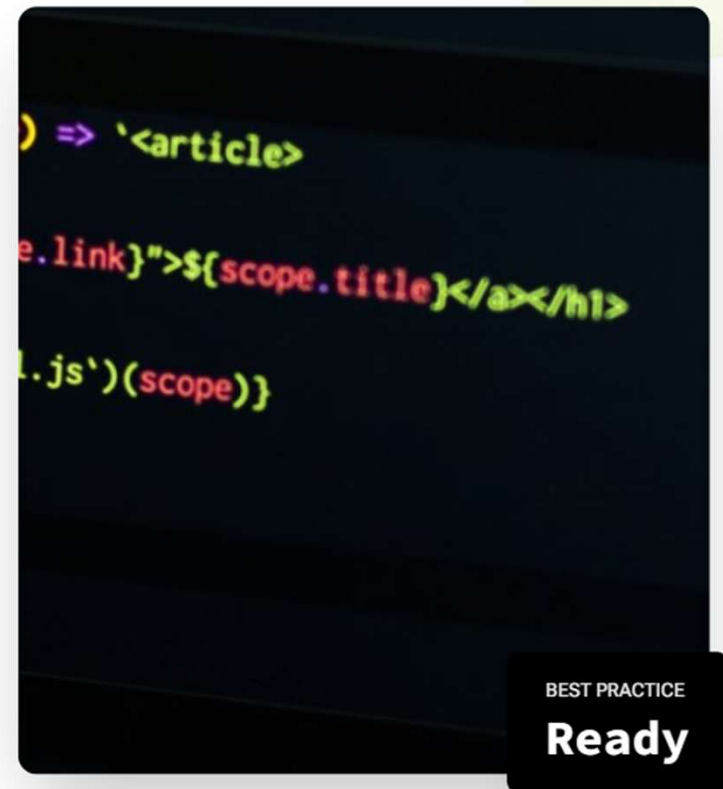
```
$(document).ready(function(){  
    // jQuery code  
});
```

02

## Short Version

RECOMMENDED

```
$(function(){  
    // code  
});
```



# | jQuery Selectors

Selectors are used to **select HTML elements** for manipulation

## ● Element

```
$("p")
```

Select all paragraphs

## ● ID

```
$("#id")
```

Select element by id

## ● Class

```
$(".class")
```

Select elements by class

## ● Universal

```
$("*")
```

Select all elements

## <> Example

```
$("#title").css("color","
```

Sets the **text color** of the element with  
**id="title"** to blue

### 💡 TIP

ID selectors are the **most efficient** and should be preferred when possible

# | jQuery Events

Events occur when **users interact** with the page

 **click()**  
Mouse click

 **dblclick()**  
Double click

 **mouseenter()**  
Mouse enters

 **mouseleave()**  
Mouse leaves

 **keypress()**  
Key pressed

 **submit()**  
Form submission

## <> Example

```
$("#button").click(function(){  
    alert("Button clicked!");  
});
```

Displays an alert when the button is **clicked**

### 💡 TIP

Events can be attached to **multiple elements** at once using selectors

# | jQuery Effects

## ● Hide & Show

```
$("#box").hide();
```

```
$("#box").show();
```

## ● Toggle

```
$("#box").toggle();
```

## ● Fade Effects

```
$("#box").fadeIn();
```

```
$("#box").fadeOut();
```

## ● Slide Effects

```
$("#box").slideDown();
```

```
$("#box").slideUp();
```



## ✦ Built-in Animations

jQuery provides **smooth, cross-browser** animation effects with minimal code

### 💡 TIP

All effects support **duration parameters** (fast, slow, milliseconds)

# | jQuery DOM Manipulation

DOM manipulation means **changing HTML content** dynamically

## ● Changing Text

```
$("#demo").text("Hello World");
```

## ● Changing HTML

```
$("#demo").html("<b>Hello</b>");
```

## ● Changing CSS

```
$("#demo").css("color", "red");
```



## ✦ Powerful Methods

DOM manipulation is **quick and easy** with jQuery's built-in methods

### 💡 TIP

Use `.html()` for HTML tags and `.text()` for plain text

# jQuery Add and Remove Elements



## Add Content

Append to element

```
$("#demo").append("New Text");
```

Adds content **at the end** of selected element



## Remove Element

Delete from DOM

```
$("#demo").remove();
```

Deletes the selected element **entirely** from the document



## Dynamic DOM

Easily **add** new content or **remove** elements dynamically

### TIP

Use `.empty()` to clear only the content, keeping the element

# | jQuery AJAX

AJAX allows **data loading** without refreshing the page

## <> AJAX Example

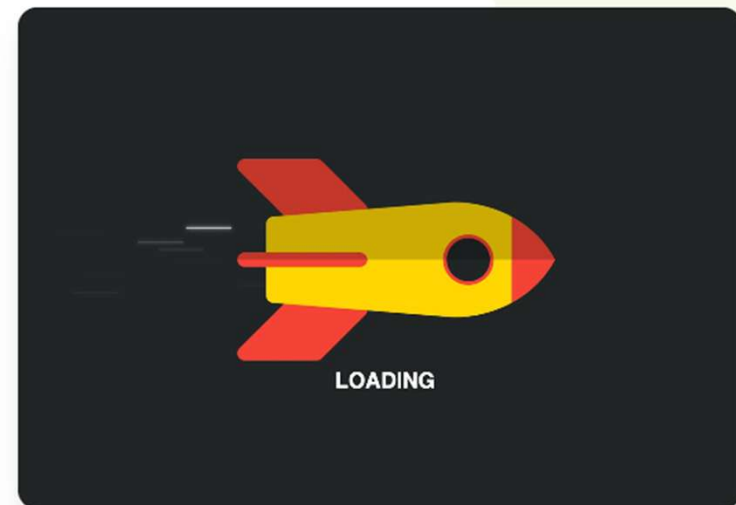
```
$.ajax({  
  url: "data.txt",  
  success: function(result){  
    $("#demo").html(result);  
  }  
});
```

### 📌 No Refresh

Updates content dynamically

### 🚀 Fast Response

Quick data retrieval



## ⇄ Asynchronous Data

Load and display data **without page reload**

### 💡 TIP

Use **.load()** for simpler server requests

# | Advantages of jQuery



## Easy to Learn

Simple syntax and intuitive API



## Less Code

Write less, achieve more functionality



## Cross-browser Compatibility

Works consistently across all browsers



## Powerful Animations

Built-in effects and transitions



## Simplifies JavaScript Tasks

Makes complex tasks easier



### ✓ Why jQuery?

jQuery is the **most popular** JavaScript library, making web development faster and easier

MARKET SHARE

**77%**



# | Disadvantages of jQuery



## Slower than pure JavaScript

jQuery adds abstraction layer, which can impact performance in **high-performance applications**



## Not required in modern frameworks

Frameworks like **React, Vue, Angular** have built-in DOM manipulation, making jQuery unnecessary



## Adds extra library size

jQuery minified size is **~30KB**, which increases page load time, especially on mobile devices



### When to Consider

These disadvantages matter most in **modern web apps** and **mobile-first** projects

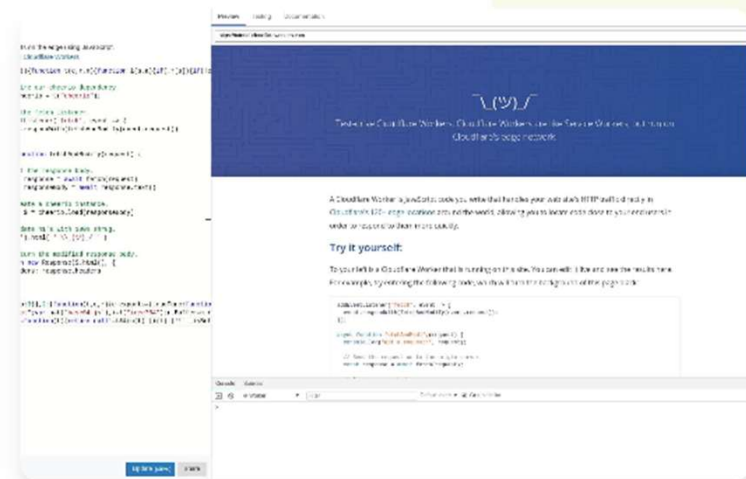
### TIP

Still useful for **legacy projects** and **simple websites**

# | jQuery vs JavaScript

| Feature               | JavaScript     | jQuery          |
|-----------------------|----------------|-----------------|
| Code Length           | Longer         | Shorter         |
| Learning              | Harder         | Easier          |
| Browser Compatibility | Needs handling | Built-in        |
| Speed                 | Faster         | Slightly slower |

 **JavaScript** offers better performance, while **jQuery** provides easier development



## ➡ Key Difference

jQuery is a **library built on JavaScript**, not a replacement

## 💡 RECOMMENDATION

Use **jQuery** for rapid development, **JavaScript** for performance

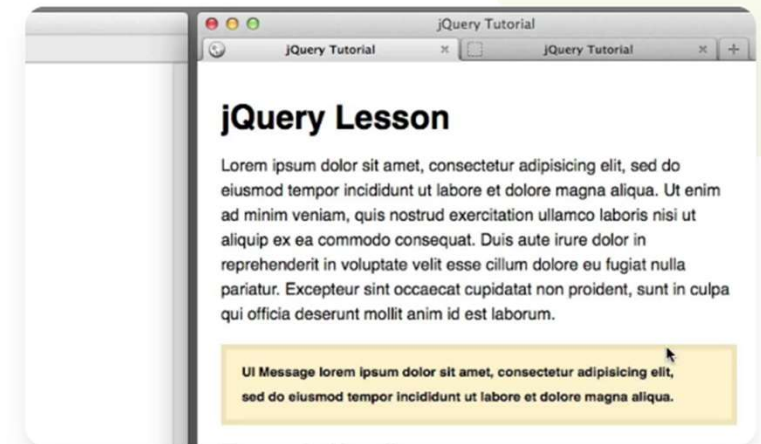
# Example Complete jQuery Program

## 1 HTML

```
<button id="btn">Click Me</button>
<p id="text">Hello</p>
```

## 2 jQuery

```
$(document).ready(function(){
    $("#btn").click(function(){
        $("#text").hide();
    });
});
```



### How It Works

When you **click the button**, the paragraph **disappears**

### KEY CONCEPTS

- Document ready event
- Button click handler
- Element selection
- Hide effect

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